Abdullah Naveed

Ph.D. Student in Computer Engineering | University of Florida

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Research Interests

Dependable AI and large language model resilience; fault tolerance and reliability for high-performance computing; scientific data compression; hardware–software co-design; machine learning systems.

Education

University of Florida **Aug 2025 – Present**
Ph.D. in Computer Engineering, GPA: 3.93/4.00 *Gainesville, FL*

Advisor: Prof. Guanpeng Li | *Research:* Dependable AI and HPC systems, fault tolerance, and scientific data compression.

University of Iowa **Aug 2023 – Aug 2025**
Master of Computer Science, GPA: 3.99/4.00 *Iowa City, IA*

Selected graduate coursework: Compiler Optimization for HPC; Distributed Systems and Algorithms; Applied Machine Learning; Functional Programming and Algorithm Design; Data Visualization and Technologies.

Lahore University of Management Sciences **Aug 2019 – May 2023**
B.S. in Computer Science, with Distinction, GPA: 3.61/4.00 *Lahore, Pakistan*

Research Experience

Argonne National Laboratory **Jun 2025 – Present**
Visiting Graduate Researcher *Lemont, IL*

- Research hardware–software co-designed, error-bounded compression for high-rate scientific data pipelines at the Advanced Photon Source, with emphasis on reducing data-movement bottlenecks through FPGA acceleration.
- Design and validate reusable Chisel/RTL compression components and streaming datapaths for composable FPGA pipelines, evaluating throughput, numerical fidelity, timing, and resource efficiency.

University of Florida **Aug 2025 – Present**
Graduate Research Assistant, Dependable Systems Lab *Gainesville, FL*

- Characterize the resilience of LLM inference, pretraining, and parameter-efficient fine-tuning to transient hardware errors using large-scale, controlled fault-injection campaigns across model architectures, workloads, and fault conditions.
- Develop low-overhead techniques for detecting and mitigating silent errors by exploiting fault-propagation behavior and informative internal model signals while preserving execution efficiency.

University of Iowa **Aug 2023 – Aug 2025**
Research Assistant *Iowa City, IA*

- Developed LLVM-based fault-injection and analysis tooling to characterize how transient errors propagate through HPC applications and lead to silent errors.
- Conducted a comparative analysis of compiler/runtime error-detection techniques across scientific workloads, examining their behavior under different fault-propagation patterns and quantifying trade-offs between detection coverage and runtime overhead.

Publications

1. **Abdullah Naveed**, Mahina Sheikh, and Guanpeng Li.
   "Low-Overhead Silent Error Detection for Large Language Model Inference at Scale."
   International Conference for High Performance Computing, Networking, Storage and Analysis (SC '26), 2026.
   Accepted; to appear.
2. Darren Ng, **Abdullah Naveed**, Duo Zhang, Guanpeng Li, Sheng Di, and Xiaoyi Lu.
   "Z-SPDK: Accelerating Lossy Compressed I/O on NVMe-SSDs."
   International Conference for High Performance Computing, Networking, Storage and Analysis (SC '26), 2026.
   Accepted; to appear.
3. Muhammad Hassan, Talha Tahir, Muhammad Farrukh, **Abdullah Naveed**, Anas Naeem, Fareed Zaffar, Fahad Shaon, Ashish Gehani, and Sazzadur Rahaman.
   "Evaluating Container Debloaters."
   2023 IEEE Secure Development Conference (SecDev), pp. 88–98, 2023. doi:10.1109/SecDev56634.2023.00023.

Manuscripts Under Review

1. **LLM Pretraining Reliability**
   Large-scale characterization of transient soft-error effects and resilience during Transformer pretraining.
   Manuscript under review, 2026.
2. **LLM Fault Resilience**
   Large-scale study of LLM behavior under increasing fault intensity, multi-bit corruption, and emerging high-error computing environments.
   Manuscript under review, 2026.
3. **HPC Reliability**
   Compiler-based, low-overhead detection of silent errors in scientific and HPC applications.
   Manuscript under review, 2026.

Teaching Experience

University of Iowa — **Aug 2023 – Dec 2023**
Teaching Assistant, Database Systems — *Iowa City, IA*

- Guided 30+ students in applying relational database concepts to MySQL projects, including schema design, query formulation, debugging, and implementation decisions.

Lahore University of Management Sciences — **Sep 2022 – Jan 2023**
Teaching Assistant, Data Structures — *Lahore, Pakistan*

- Led tutorials for 100+ students on data structures and algorithms, reinforcing core concepts through worked examples and guiding students in applying them to programming and problem-solving assignments.

Selected Technical Projects

Multimodal Fake News Classification
Developed a multimodal classification pipeline that integrates textual and visual representations to identify misleading content using complementary evidence across language and image modalities.
Technologies: Python, NLP, Computer Vision, Multimodal Learning

Cozmo's Star Wars Odyssey
Built an interactive robotics experience combining visual character recognition with LLM-driven narrative generation and question answering to enable context-aware interaction with a Cozmo robot.
Technologies: Computer Vision, Large Language Models, Robotics

COVID SOP Violation Detection
Developed a real-time multi-camera vision pipeline for mask detection and crowd monitoring, integrating YOLOv5-based detection, top-view projection, and spatial heatmaps for density analysis.
Technologies: Python, PyTorch, OpenCV, YOLOv5

Audio Sentiment Analysis
Developed a speech-emotion classification pipeline using 7,442 labeled CREMA-D audio clips to learn and evaluate affective patterns in spoken audio.
Technologies: Python, Speech Processing, Machine Learning

Netflix Recommendation System
Built a personalized content-recommendation pipeline using NLP-based content representations and unsupervised clustering to model content similarity and generate user-oriented recommendations.
Technologies: Python, scikit-learn, NLP, Unsupervised Learning

Technical Skills

Programming: Python, C, C++, Bash, SQL, Scala/Chisel, JavaScript, Git.
Machine Learning / AI: PyTorch, Hugging Face Transformers, TensorFlow/Keras, scikit-learn, OpenCV.
HPC / Parallel Computing: MPI, OpenMP, Slurm, large-scale CPU/GPU experimentation, fault injection, resilience analysis.
Hardware / Systems: Chisel, Verilog/RTL, Chipyard, RISC-V, Verilator, Vivado, LLVM, Linux, Docker.
Scientific Computing / Data Analysis: NumPy, Pandas, Matplotlib, Jupyter, MATLAB.

Professional Service

Conference Reviewer / Subreviewer: SC; DSN; ICS; ISRAE.

Research Community Contribution: Contributing team member to the ProdeBench benchmarking project, supporting implementation and experimental evaluation.